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Power System Planning and Reliability (PSM-104): Module-wise Question Bank

*Semester Question Bank Analysis
with Course Outcomes and Bloom's Taxonomy Levels*

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1 Introduction

This document presents a comprehensive analysis and organization of End Semester question papers for the course **Power System Planning and Reliability (PSM-104)**. The questions have been systematically classified according to:

- **Module Number:** As per the course syllabus structure
- **Question Type:** MCQ (1 mark), Short Answer (5 marks), Long Answer (15 marks)
- **Course Outcomes (CO):** Mapped to PSM-104.1 through PSM-104.5
- **Bloom's Taxonomy (BT) Levels:** As per NEP 2020 guidelines

1.1 Bloom's Taxonomy Levels (NEP 2020)

The cognitive levels are classified as follows:

BT-1 (Remember): Recall facts, terms, basic concepts

BT-2 (Understand): Explain ideas, interpret information, summarize concepts

BT-3 (Apply): Use information in new situations, solve problems

BT-4 (Analyze): Draw connections, examine relationships, differentiate

BT-5 (Evaluate): Justify decisions, critique approaches, assess alternatives

BT-6 (Create): Design, construct, develop new solutions or systems

1.2 Course Outcomes

PSM-104.1: Implement reliability-based planning criteria and risk analysis

PSM-104.2: Apply security analysis methodologies and approaches

PSM-104.3: Develop distribution systems and optimize protection coordination

PSM-104.4: Utilize advanced reliability evaluation techniques and maintenance approaches

PSM-104.5: Development of network configuration strategies

2 Multiple Choice Questions (1 Mark Each)

This section contains all MCQ questions organized by module, with corresponding Course Outcomes and Bloom's Taxonomy levels.

2.1 Module 1: System Planning

Problem 2.1:

The primary objective of long-term power system planning typically ranges over a time horizon of:

CO: PSM-104.1 **BT Level:** BT-1 (Remember)

Problem 2.2:

The reserve margin in short-term generation planning is primarily maintained to account for:

CO: PSM-104.1 **BT Level:** BT-2 (Understand)

Problem 2.3:

Reactive power planning is primarily concerned with:

CO: PSM-104.1 **BT Level:** BT-2 (Understand)

Problem 2.4:

Medium-term power system planning typically addresses a time horizon of _____ years.

CO: PSM-104.1 **BT Level:** BT-1 (Remember)

Problem 2.5:

The primary factor that differentiates short-term planning from long-term planning is the _____ horizon.

CO: PSM-104.1 **BT Level:** BT-2 (Understand)

Problem 2.6:

In power system planning, the time horizon for short-term planning typically ranges from _____ months to 2 years.

CO: PSM-104.1 **BT Level:** BT-1 (Remember)

Problem 2.7:

The main objective of reactive power planning is to maintain acceptable _____ levels throughout the power system.

CO: PSM-104.1 **BT Level:** BT-2 (Understand)

Problem 2.8:

Environmental considerations in power system planning include assessment of _____ emissions and their mitigation.

CO: PSM-104.1 **BT Level:** BT-2 (Understand)

Problem 2.9:

Load duration curve is a graphical representation of load arranged in _____ order of magnitude.

CO: PSM-104.1 **BT Level:** BT-1 (Remember)

2.2 Module 2: Reliability

Problem 2.10:

Which probabilistic index represents the expected frequency of load curtailment events per year?

CO: PSM-104.2 **BT Level:** BT-2 (Understand)

Problem 2.11:

The availability of a component with MTTF of 2000 hours and MTTR of 50 hours is _____ %.

CO: PSM-104.1 **BT Level:** BT-3 (Apply)

Problem 2.12:

In the Markov process for reliability analysis, the steady-state probability is obtained when the system reaches _____ condition.

CO: PSM-104.2 **BT Level:** BT-2 (Understand)

Problem 2.13:

The recursive technique for reliability evaluation is particularly useful for systems with _____ number of components.

CO: PSM-104.2 **BT Level:** BT-2 (Understand)

Problem 2.14:

The ratio of Mean Time To Failure to the sum of MTTF and MTTR gives the component _____.

CO: PSM-104.1 **BT Level:** BT-3 (Apply)

2.3 Module 3: Generation Planning and Reliability

Problem 2.15:

In the context of generation reliability, the forced outage rate (FOR) of a generating unit is calculated as:

CO: PSM-104.2 **BT Level:** BT-3 (Apply)

Problem 2.16:

The capacity credit of a renewable generation source indicates:

CO: PSM-104.2 **BT Level:** BT-2 (Understand)

Problem 2.17:

Which method is commonly used for medium-term load forecasting?

CO: PSM-104.2 **BT Level:** BT-1 (Remember)

Problem 2.18:

In generation expansion planning, the Loss of Load Expectation (LOLE) is typically expressed in _____ per year.

CO: PSM-104.2 **BT Level:** BT-1 (Remember)

Problem 2.19:

The capacity outage probability table is primarily used in _____ reliability evaluation.

CO: PSM-104.2 **BT Level:** BT-2 (Understand)

Problem 2.20:

Scheduled outage of generating units is planned during _____ demand periods to minimize system impact.

CO: PSM-104.2 **BT Level:** BT-2 (Understand)

2.4 Module 4: Transmission Planning and Reliability

Problem 2.21:

The N-1 security criterion in transmission planning ensures that the system can withstand:

CO: PSM-104.2 **BT Level:** BT-2 (Understand)

Problem 2.22:

In transmission planning, composite system reliability assessment considers both _____ and transmission system failures.

CO: PSM-104.2 **BT Level:** BT-2 (Understand)

Problem 2.23:

In transmission planning, the transfer capability between two areas is limited by _____ and voltage constraints.

CO: PSM-104.2 **BT Level:** BT-2 (Understand)

2.5 Module 5: Distribution Planning and Reliability

Problem 2.24:

System Average Interruption Duration Index (SAIDI) is measured in:

CO: PSM-104.3 **BT Level:** BT-1 (Remember)

Problem 2.25:

In distribution system reliability analysis, SAIFI represents:

CO: PSM-104.3 **BT Level:** BT-1 (Remember)

Problem 2.26:

Customer Average Interruption Duration Index (CAIDI) is calculated by dividing _____ by SAIFI.

CO: PSM-104.3 **BT Level:** BT-3 (Apply)

Problem 2.27:

The optimal placement of shunt capacitors in distribution systems primarily aims to improve _____ profile and reduce losses.

CO: PSM-104.3 **BT Level:** BT-2 (Understand)

Problem 2.28:

Expected Energy Not Served (EENS) is measured in units of _____ per year.

CO: PSM-104.3 **BT Level:** BT-1 (Remember)

Problem 2.29:

Bus bar failure in distribution systems is classified as a _____ type of failure.

CO: PSM-104.3 **BT Level:** BT-1 (Remember)

Problem 2.30:

The effect of transferring loads in distribution systems can improve _____ by providing alternate supply paths.

CO: PSM-104.3 **BT Level:** BT-2 (Understand)



3 SHORT ANSWER TYPE QUESTIONS

This section contains short answer type questions (5 marks each) organized by module.

3.1 Module 1: System Planning

Problem 3.1:

Explain the significance of planning reserve margin in generation adequacy assessment. Derive the relationship between installed capacity, peak demand, and reserve margin. What are typical reserve margin values adopted in Indian power systems?

CO: PSM-104.1, PSM-104.2 **BT Level:** BT-2, BT-3

Problem 3.2:

Explain the significance of load forecasting in power system planning. Compare and contrast at least two load forecasting techniques used in medium-term planning. What are the key factors influencing load forecast accuracy?

CO: PSM-104.1 **BT Level:** BT-2, BT-4

3.2 Module 2: Reliability

Problem 3.3:

Differentiate between deterministic and probabilistic planning approaches in power systems. Discuss the advantages of probabilistic methods with a suitable example related to generation capacity planning.

CO: PSM-104.1, PSM-104.2 **BT Level:** BT-4

Problem 3.4:

Explain the concept of stochastic prediction in power system reliability analysis. How does it differ from deterministic approaches? Discuss the application of stochastic methods in predicting frequency and duration of interruptions.

CO: PSM-104.1, PSM-104.4 **BT Level:** BT-2, BT-4

Problem 3.5:

Describe the recursive technique for calculating system reliability. Illustrate the method with a simple example of a three-component series system showing the step-by-step calculation procedure.

CO: PSM-104.4 **BT Level:** BT-2, BT-3

3.3 Module 3: Generation Planning and Reliability

Problem 3.6:

Describe the concept of Loss of Load Probability (LOLP) in generation system planning. Outline the computational procedure for calculating LOLP using the recursive technique.

CO: PSM-104.2, PSM-104.4 **BT Level:** BT-2, BT-3

Problem 3.7:

What is Integrated Resource Planning (IRP)? Explain how IRP incorporates both supply-side and demand-side resources in generation planning. Discuss the advantages of IRP over traditional planning methods.

CO: PSM-104.2 **BT Level:** BT-2, BT-4

Problem 3.8:

Elaborate on the factors affecting interconnection between power systems under emergency assistance. What are the reliability benefits and challenges of interconnected system operation?

CO: PSM-104.2, PSM-104.4 **BT Level:** BT-2, BT-4

Problem 3.9:

Describe the concept of Loss of Energy (LOE) in generation system reliability. How does it differ from Loss of Load Probability? Explain the evaluation methods for calculating LOE in interconnected systems.

CO: PSM-104.2, PSM-104.4 **BT Level:** BT-2, BT-4

Problem 3.10:

Explain the concept of capacity expansion in generation planning. What are the key decision variables and constraints in the capacity expansion optimization problem? Discuss the role of lead time in expansion planning.

CO: PSM-104.2 **BT Level:** BT-2, BT-4

3.4 Module 4: Transmission Planning and Reliability

Problem 3.11:

Discuss the factors affecting reactive power planning in transmission systems. Explain how voltage stability margin is quantified and its importance in system security assessment.

CO: PSM-104.2, PSM-104.3 **BT Level:** BT-2, BT-4

Problem 3.12:

Explain the role of network reconfiguration in enhancing transmission system reliability. Describe at least three strategies for improving load point reliability indices through transmission network optimization.

CO: PSM-104.2, PSM-104.3, PSM-104.5 **BT Level:** BT-2, BT-3

Problem 3.13:

Explain the concept of load point indices in transmission system reliability assessment. Describe how system indices differ from load point indices and their significance in planning studies.

CO: PSM-104.2, PSM-104.3 **BT Level:** BT-2, BT-4

Problem 3.14:

Describe the network reconfiguration process in transmission systems. What are the primary objectives of reconfiguration? Explain how reliability indices are affected by different network topologies.

CO: PSM-104.2, PSM-104.3, PSM-104.5 **BT Level:** BT-2, BT-4

3.5 Module 5: Distribution Planning and Reliability

Problem 3.15:

Define the following distribution system reliability indices and explain their significance: (i) SAIFI, (ii) SAIDI, (iii) CAIDI. How are these indices mathematically related?

CO: PSM-104.3, PSM-104.4 **BT Level:** BT-2, BT-3

Problem 3.16:

Discuss the effects of lateral distribution protection on system reliability. How do fuses, reclosers, and sectionalizers contribute to improving distribution system performance?

CO: PSM-104.3, PSM-104.4 **BT Level:** BT-2, BT-4

Problem 3.17:

Discuss the effects of protection failure on distribution system reliability indices. How do primary and backup protection coordination strategies minimize the impact of protection failures?

CO: PSM-104.3, PSM-104.4 **BT Level:** BT-2, BT-4

Problem 3.18:

Discuss the concept of breaker failure in distribution systems and its impact on customer reliability. Explain the protection schemes used to mitigate the effects of breaker failure.

CO: PSM-104.3, PSM-104.4 **BT Level:** BT-2, BT-4

4 LONG ANSWER TYPE QUESTIONS

This section contains long answer type questions (15 marks each) organized by module.

4.1 Module 1: System Planning

Problem 4.1:

Part (a): Elaborate on the hierarchical structure of power system planning across different time horizons (short-term, medium-term, and long-term). What are the key objectives and constraints specific to each planning horizon?

Part (b): A power system has a current peak demand of 5000 MW growing at 4% annually. The system has 6200 MW installed capacity with an average forced outage rate of 8%. Calculate:

- (i) Current planning reserve margin
- (ii) Available capacity considering forced outages
- (iii) Required capacity addition for maintaining 15% reserve margin after 3 years

Part (c): Discuss how renewable energy integration affects traditional generation planning methodologies.

CO: PSM-104.1, PSM-104.2 **BT Level:** BT-2, BT-3, BT-4

Problem 4.2:

Part (a): Elaborate on the factors affecting power system planning decisions. Discuss the role of economic, technical, environmental, and regulatory constraints in the planning process. How do these factors interact in modern planning studies?

Part (b): A utility forecasts the following load growth over the next 5 years:

- Current year peak load: 4200 MW
- Annual growth rate: Years 1-3 at 3.5%, Years 4-5 at 4.2%
- Current installed capacity: 5100 MW
- Planned retirements: 300 MW in Year 2, 250 MW in Year 4
- Target reserve margin: 20%

Calculate:

- (i) Peak load for each year
- (ii) Required installed capacity for each year
- (iii) Capacity additions needed in each year
- (iv) Total new capacity requirement over 5 years

Part (c): Explain how demand-side management programs affect generation planning. Discuss at least three DSM strategies and their impact on capacity requirements.

CO: PSM-104.1, PSM-104.2 **BT Level:** BT-2, BT-3, BT-4, BT-5

4.2 Module 2: Reliability - Markov Process

Problem 4.3:

Part (a): Discuss the Markov process approach for power system reliability evaluation. Develop the state transition diagram for a two-component system and derive expressions for steady-state probabilities.

CO: PSM-104.4 **BT Level:** BT-2, BT-3, BT-4

Problem 4.4:

Part (a): Derive the fundamental equations for the Markov process approach in reliability evaluation. Explain the concept of state space representation and transition rates between different system states.

Part (b): A two-component system has the following data:

- Component 1: Failure rate = 0.02 failures/year, Repair rate = 200 repairs/year
- Component 2: Failure rate = 0.03 failures/year, Repair rate = 150 repairs/year
- Both components operate in parallel

Develop the state transition diagram and calculate:

- (i) Steady-state probabilities for all states
- (ii) System availability
- (iii) Expected frequency of system failure

Part (c): Discuss how the Markov model can be extended for multi-component systems and the computational challenges involved.

CO: PSM-104.4 **BT Level:** BT-2, BT-3, BT-4

4.3 Module 3: Generation Planning and Reliability

Problem 4.5:

Part (a): Derive the mathematical formulation for the least-cost generation expansion planning problem. Clearly state the objective function, decision variables, and major constraints.

Part (b): Explain the concept of Integrated Resource Planning (IRP) in modern power systems. How does IRP differ from traditional supply-side planning approaches?

Part (c): A utility is planning capacity additions for a 10-year horizon. Three technology options are available:

- Coal: Capital cost Rs. 50,000/kW, Variable cost Rs. 2.5/kWh, Capacity factor 80%
- Gas: Capital cost Rs. 35,000/kW, Variable cost Rs. 4.0/kWh, Capacity factor 60%
- Solar: Capital cost Rs. 40,000/kW, Variable cost Rs. 0/kWh, Capacity factor 20%

Using a discount rate of 8%, determine the levelized cost of electricity for each technology. Comment on the economic competitiveness.

CO: PSM-104.2 **BT Level:** BT-3, BT-4, BT-5

Problem 4.6:

Part (a): Explain the concept of adequacy of reliability in power systems. Differentiate between adequacy and security aspects of system reliability. Why is adequacy assessment crucial in planning studies?

Part (b): A power system serves a peak load of 3500 MW with the following generation mix:

- Unit Type A: 5 units $\times$ 400 MW each, FOR = 5%
- Unit Type B: 8 units $\times$ 250 MW each, FOR = 8%
- Unit Type C: 10 units $\times$ 100 MW each, FOR = 12%

Calculate:

- (i) Total installed capacity
- (ii) Expected available capacity
- (iii) Planning reserve margin
- (iv) Is the system adequate if minimum reserve margin requirement is 18%?

Part (c): Discuss the relationship between reliability cost and system reliability level. How does cost-benefit analysis influence planning decisions?

CO: PSM-104.1, PSM-104.2 **BT Level:** BT-2, BT-3, BT-4, BT-5

Problem 4.7:

A generation planning study evaluates three scenarios for capacity addition:

- Scenario 1: Add 800 MW coal plant, Capital cost Rs. 3.2 crores/MW, Variable cost Rs. 2.8/kWh, FOR = 6%
- Scenario 2: Add 600 MW gas plant, Capital cost Rs. 2.4 crores/MW, Variable cost Rs. 4.5/kWh, FOR = 4%
- Scenario 3: Add 1000 MW solar + 200 MW storage, Capital cost Rs. 5.0 crores/MW, Variable cost Rs. 0.5/kWh, FOR = 2%
- Plant load factor: Coal 75%, Gas 55%, Solar 22%

Using 25-year life and 9% discount rate, determine:

- (i) Levelized cost for each scenario
- (ii) Expected available capacity considering FOR
- (iii) Most economical option

CO: PSM-104.2 **BT Level:** BT-3, BT-4, BT-5

Problem 4.8:

Part (a): Describe the process of generation system modeling for reliability studies. Explain the significance of capacity outage probability tables and their construction. How are load models integrated with generation models?

Part (b): A generation system has the following units:

- 3 units $\times$ 500 MW, FOR = 4%

- 5 units $\times$ 300 MW, FOR = 6%
- 6 units $\times$ 200 MW, FOR = 9%
- System peak load: 3800 MW with a load forecast uncertainty of $\pm 5\%$

Using capacity outage probability approach, calculate:

- Total installed capacity
- Expected capacity on outage
- Reserve margin percentage
- Probability that available capacity is less than peak load (simplified calculation assuming independent failures)

Part (c): Explain the concept of spinning reserve and its importance in generation reliability. How does spinning reserve differ from total reserve? Discuss optimal spinning reserve determination.

CO: PSM-104.2, PSM-104.4 **BT Level:** BT-2, BT-3, BT-4

4.4 Module 4: Transmission Planning and Reliability

Problem 4.9:

Part (a): Describe the transmission system planning process with emphasis on network reconfiguration strategies for reliability enhancement. What are the key data requirements for composite system reliability assessment?

Part (b): Explain the following concepts in reactive power planning:

- Voltage control hierarchy
- Optimal VAR compensation placement
- Voltage stability margin assessment
- Coordination between generator excitation systems, tap changers, and shunt compensators

Part (c): A 230 kV transmission corridor has three alternative routing options:

- Route 1: Failure rate 0.5/year, Capital cost Rs. 800 crores
- Route 2: Failure rate 0.3/year, Capital cost Rs. 1100 crores
- Route 3: Failure rate 0.2/year, Capital cost Rs. 1400 crores
- Average repair time for all routes: 24 hours
- Annual load served: 1500 MW
- Interruption cost: Rs. 15,000/MWh

Using a 20-year planning horizon and 10% discount rate, determine which route is economically optimal considering both investment and reliability costs.

CO: PSM-104.2, PSM-104.3, PSM-104.5 **BT Level:** BT-2, BT-3, BT-4, BT-5

Problem 4.10:

Discuss the objectives of reactive power planning in power systems. Explain the coordination mechanism between different reactive power sources including generators, synchronous condensers, static VAR compensators, and capacitor banks.

CO: PSM-104.2, PSM-104.3 **BT Level:** BT-2, BT-4

Problem 4.11:

A transmission project evaluation involves comparing two alternatives:

- Alternative A: Overhead line
 - Capital cost: Rs. 1200 crores
 - Annual O&M: Rs. 18 crores
 - Failure rate: 0.4 f/yr
 - Repair time: 30 hrs
- Alternative B: Underground cable
 - Capital cost: Rs. 2800 crores
 - Annual O&M: Rs. 12 crores
 - Failure rate: 0.08 f/yr
 - Repair time: 120 hrs
- Average load served: 2000 MW
- Interruption cost: Rs. 12,000/MWh
- Study period: 30 years, Discount rate: 8%

Calculate for each alternative:

- (i) Annual reliability cost
- (ii) Present worth of total costs
- (iii) Levelized annual cost
- (iv) Recommend the better alternative with justification

CO: PSM-104.2, PSM-104.3 **BT Level:** BT-3, BT-4, BT-5

4.5 Module 5: Distribution Planning and Reliability

Problem 4.12:

A distribution feeder serves 5000 customers with the following annual outage data:

- Component A: Failure rate = 0.8 failures/year, Repair time = 3 hours
- Component B: Failure rate = 1.2 failures/year, Repair time = 2 hours
- Component C: Failure rate = 0.5 failures/year, Repair time = 4 hours
- All components in series configuration

Calculate SAIFI, SAIDI, and CAIDI for this feeder. Explain how lateral distribution protection devices affect system reliability indices. What is the impact of automated switching on customer interruption statistics?

CO: PSM-104.3, PSM-104.4 **BT Level:** BT-3, BT-4

Problem 4.13:

Part (a): Describe the data requirements for composite system (generation and transmission) reliability assessment. Explain the significance of each data element in reliability evaluation studies.

Part (b): A distribution network serves 8000 customers through a radial feeder with four protective devices in series:

- Device A: Failure rate = 0.6 failures/year, Switching time = 1.5 hours
- Device B: Failure rate = 0.9 failures/year, Switching time = 2.0 hours
- Device C: Failure rate = 0.4 failures/year, Switching time = 1.0 hours
- Device D: Failure rate = 1.1 failures/year, Switching time = 2.5 hours
- 30% of customers are served beyond each device

Calculate the system SAIFI and SAIDI. If disconnects are installed to isolate 50% of customers during faults, recalculate the indices.

Part (c): Explain the impact of network reconfiguration on distribution system reliability. Describe at least two reconfiguration strategies.

CO: PSM-104.2, PSM-104.3, PSM-104.4, PSM-104.5 **BT Level:** BT-2, BT-3, BT-4

Problem 4.14:

Explain how scheduled maintenance and temporary failures affect generation system reliability. Discuss maintenance planning strategies to minimize reliability impact.

CO: PSM-104.2, PSM-104.4 **BT Level:** BT-2, BT-4

Problem 4.15:

Part (a): Discuss the concept of frequency and duration of interruptions in power system reliability. Derive the relationship between failure rate, repair rate, and expected interruption frequency for a single component.

Part (b): A radial distribution system has the following series-connected components serving 6000 customers:

- Section 1: $\lambda_1 = 0.5$ f/yr, $r_1 = 4$ hrs, serves 100% customers
- Section 2: $\lambda_2 = 0.7$ f/yr, $r_2 = 3$ hrs, serves 75% customers
- Section 3: $\lambda_3 = 0.4$ f/yr, $r_3 = 5$ hrs, serves 50% customers
- Section 4: $\lambda_4 = 0.9$ f/yr, $r_4 = 2$ hrs, serves 25% customers

Calculate:

- (i) SAIFI for the system
- (ii) SAIDI for the system
- (iii) CAIDI for the system

(iv) ASAI (Average Service Availability Index)

Part (c): Explain the effects of disconnects in distribution systems. How do sectionalizing switches improve reliability indices? Support with examples.

CO: PSM-104.3, PSM-104.4 **BT Level:** BT-2, BT-3, BT-4

Problem 4.16:

Part (a): Discuss parallel and meshed network configurations in distribution systems. Compare their reliability performance with radial networks. What are the operational and economic trade-offs in selecting network topology?

Part (b): Explain the concept of temporary and transient failures in power systems. How do automatic reclosing schemes improve system reliability? Discuss the impact on reliability indices.

CO: PSM-104.3, PSM-104.4, PSM-104.5 **BT Level:** BT-2, BT-4, BT-5

5 Summary Statistics

5.1 Question Distribution by Module

Table 1: Distribution of Questions Across Modules

Module	MCQ (1 mark)	Short (5 marks)	Long (15 marks)
Module 1: System Planning	9	2	2
Module 2: Reliability	5	3	2
Module 3: Generation Planning	6	5	5
Module 4: Transmission Planning	3	4	4
Module 5: Distribution Planning	7	4	7
Total	30	18	20

5.2 Bloom's Taxonomy Distribution

Table 2: Distribution of Questions by Bloom's Taxonomy Level

BT Level	MCQ	Short	Long
BT-1 (Remember)	11 (37%)	–	–
BT-2 (Understand)	16 (53%)	15 (83%)	18 (90%)
BT-3 (Apply)	3 (10%)	8 (44%)	17 (85%)
BT-4 (Analyze)	–	13 (72%)	19 (95%)
BT-5 (Evaluate)	–	–	8 (40%)
BT-6 (Create)	–	–	–

5.3 Course Outcome Coverage

Table 3: Question Count by Course Outcome

Course Outcome	MCQ	Short	Long
PSM-104.1	9	4	4
PSM-104.2	12	11	15
PSM-104.3	7	6	11
PSM-104.4	2	7	10
PSM-104.5	0	2	4

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